Problem 3

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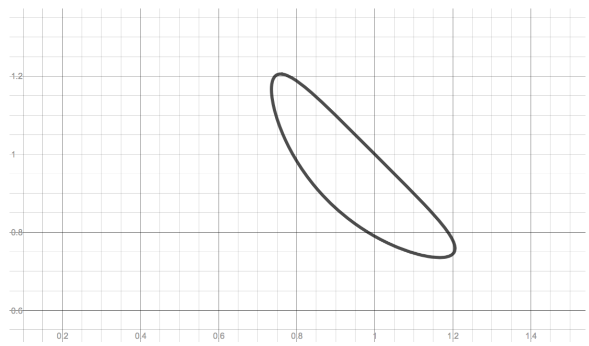
Problem 1 *A part of the curve with equation* $\cos(\pi xy)+x+y=1$ *is sketched below.*



(a) *Use the implicit differentiation to find the derivative* dy/dx*.*

(b) *Consider the point* $(1,1)$*. Show (algebraically) that this point lies on the curve.*

(c) *Find the equation of the line tangent to the curve at* $(1,1)$*. Draw this line in the figure above.*